

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$$(x - 2) - 4(y + 7) = 117$$
$$(x - 2) + 4(y + 7) = 442$$

The solution to the given system of equations is (x, y) . What is the value of $6(x - 2)$?

Correct Answer: 1677

Rationale

The correct answer is **1,677**. Adding the first equation to the second equation in the given system yields $(x - 2) + (x - 2) + (-4)(y + 7) + 4(y + 7) = 117 + 442$, or $2(x - 2) = 559$. Multiplying both sides of this equation by **3** yields $6(x - 2) = 1,677$. Therefore, the value of $6(x - 2)$ is **1,677**.